

TL;DR

BharatNXT Vendor Verification delivers a centralized, automated portal for Indian small businesses to onboard and verify suppliers. It streamlines compliance checks, mitigates fraud risk with trust scoring, and provides transparent, downloadable reports. Targeted at small business owners and administrators, the platform makes vendor verification quick, reliable, and scalable.

Goals

Business Goals

- Achieve 1,000 verified vendor onboardings within the first 3 months.
- Reduce average vendor verification time to under 5 minutes.
- Boost platform adoption among small businesses by at least 30% quarter-over-quarter.
- Establish BharatNXT as a trusted authority for vendor due diligence in India.

User Goals

- Quickly verify vendor legitimacy using automated checks.
- Showcase compliance and trust profiles to attract more business.
- Minimize exposure to fraudulent or non-compliant suppliers.
- Gain actionable, easy-to-understand verification reports for record-keeping.

Non-Goals

- Developing complex enterprise integrations (e.g., SAP, Oracle).
- Manual physical document verification or in-person KYC.
- International vendor checks (focus is India-only on launch).

User Stories

Small Business Buyer

- As a buyer, I want to verify a vendor's GSTIN so that I can be assured of the vendor's legitimacy.
- As a buyer, I want to see each vendor's compliance status in one place so that I can make fast, risk-aware decisions.

- As a buyer, I want downloadable trust reports so that I have audit trails for due diligence.

Small Business Vendor/Supplier

- As a vendor, I want to submit my compliance info once so that I do not have to repeat the process for every buyer.
- As a vendor, I want to showcase my verified status and positive trust score so that I can attract more business.

BharatNXT Admin

- As an admin, I need to monitor all onboarding and verification requests so that I can ensure timely completion.
- As an admin, I need to flag risky or incomplete verifications so that I can follow up or intervene as needed.
- As an admin, I want to analyze verification trends and outliers so that I can improve platform risk management.

Functional Requirements

- **Vendor Onboarding & Verification (Priority: P1)**
 - Vendor Onboarding Form: Capture GSTIN (mandatory), PAN, contact details, and supporting documents.
 - Real-Time GSTIN Format Validation: Input checks before submission.
 - Automated Government Validation: Connect to GSTIN and Udyam/MSME APIs to confirm compliance.
 - Reputation Scraping: Aggregate external reputation from Google, e-commerce, and business review sites.
 - Trust Score Generation: Compute a 100-point trust score based on compliance and reputation.
 - PDF Trust Report: Generate and allow download of a color-coded report (Green/Amber/Red ratings).
- **Admin Dashboard (Priority: P1)**
 - Pending Verifications Queue: View, filter, and manage ongoing requests.
 - Risk Flagging: Automated alerts for incomplete, inconsistent, or suspicious submissions.

- Manual Override/Review: Enable manual verification in case of API downtime or edge cases.
- **Premium Verification Modules (Priority: P2)**
 - Director/KYC Due Diligence: Integrate MCA21/ZAUBA company director info.
 - Payment Default History: Report on defaults via CRIF/Experian APIs.
- **Additional Requirements**
 - Responsive UI: Mobile-friendly with clear, accessible components.
 - Secure User Authentication: Email/OTP login for buyers, vendors, and admins.
 - Audit Logging: Comprehensive tracking of verification activities.
 - Secure Document Uploads: Encrypt all sensitive data and documents at rest and in transit.

User Experience

Entry Point & First-Time User Experience

- Users access BharatNXT Vendor Verification via the BharatNXT business portal or direct link.
- Website landing page gives a concise value proposition and “Start Verification” CTA.
- First-time users are prompted to sign up with email or mobile OTP. Short onboarding tutorial (1 step).

Core Experience

- **Step 1:** Vendor submits onboarding form with GSTIN, PAN, contact details, and documents.
 - Real-time field checks and format validations.
 - Simple, single-page input form for speed and clarity.
 - Progress bar for multi-step upload/verification.
- **Step 2:** Automated back-end checks (GSTIN/Udyam APIs, reputation scraping).
 - User sees “Verification in progress” with estimated completion time (<2 mins).

- Clear error messaging for failed validations or incomplete information.
- **Step 3:** Trust score generation and report presentation.
 - User sees traffic-light (Green/Amber/Red) score with detailed breakdown.
 - Downloadable PDF trust report and shareable link.
- **Step 4:** Admin dashboard for monitoring and manual intervention (if needed).
 - Admins view all entries, filter by status, and receive alerts for flagged-risk vendors.
 - Manual review flow triggered if APIs are unavailable; document upload evaluation.

Advanced Features & Edge Cases

- Premium users can unlock advanced director and payment history reports upon payment.
- In the event of API downtime, vendors can upload supporting documents for manual admin review.
- Automated email/SMS notifications for status changes, approvals, and expirations.

UI/UX Highlights

- High-contrast color scheme for traffic-light status.
- Clean, uncluttered interface with Material UI.
- Responsive design across devices.
- Always-visible help/support widget.
- Clear calls-to-action, minimal jargon, grade 8 readability.

Narrative

For small businesses across India, verifying the legitimacy and compliance of vendors has often meant time-consuming manual checks, sifting through paperwork, and putting trust in unverified claims. As digital transactions accelerate, this gap in due diligence exposes honest businesses to unnecessary risk, slows down commerce, and damages reputations.

BharatNXT Vendor Verification turns this process on its head. Through a quick, self-serve onboarding portal, vendors and buyers can validate compliance information—GSTIN, Udyam/MSME status, external reputation, and even director KYC details—in

just a few clicks. Our intelligent trust scoring system translates complex records into a simple, actionable verdict, while downloadable reports satisfy every audit requirement.

Admins, meanwhile, stay in control with a live view of all pending verifications, and can step in if automated systems encounter issues. With robust security, a transparent scoring model, and lightning-fast workflows, BharatNXT helps small businesses grow safely and with confidence. No more risky guesses—just trust, built in.

Success Metrics

User-Centric Metrics

- 95%+ of verifications completed in under 5 minutes.
- At least 90% of vendors achieve a valid trust score on first submission.
- Net Promoter Score (NPS) of 50+ among buyer and supplier users.
- <5% user drop-off during onboarding or verification flows.

Business Metrics

- 1,000+ vendors successfully onboarded in the first quarter post-launch.
- 30% month-over-month growth in verified users.
- Conversion rate of 10% to premium verification modules.

Technical Metrics

- API verification uptime >99% during business hours.
- Verification accuracy rate (match with official records) >95%.
- PDF report delivery time <10 seconds per request.

Tracking Plan

- Number of onboarding form submissions.
- Verification success/failure and reason codes.
- Trust report downloads and shares.
- Premium module purchase events.
- API call latency and error rates.
- Admin manual intervention cases.

Technical Considerations

Technical Needs

- **Frontend:** Created using React.js and Material UI, built for instant responsiveness and mobile device compatibility.
- **Backend:** Node.js/Express REST API serving onboarding, verification, scoring, and reporting workflows.
- **APIs/Integrations:** Live connections with GSTIN, Udyam/MSME, MCA21/ZAUBA for director checks; Puppeteer-based scraping for reputation; CRIF/Experian APIs for payment history.
- **Database:** PostgreSQL for core structured data (vendors, audits, scores), MongoDB for unstructured document uploads and storage.

Integration Points

- Government APIs (GSTIN, Udyam/MSME, MCA21/ZAUBA).
- Third-party data providers (CRIF/Experian).
- Email/SMS notification gateway.
- BharatNXT core authentication and user management.

Data Storage & Privacy

- Compliance with GDPR—encrypt all user and PDF data both in transit and at rest.
- User consents and purpose-based access controls throughout.
- Automated data deletion for expired/withdrawn verifications.
- Full audit trails for all verification actions.

Scalability & Performance

- Designed for 10,000+ concurrent onboardings.
- Stateless back-end for horizontal scalability.
- CDN-backed PDF distribution for fast downloads nationwide.

Potential Challenges

- Government API downtime: Fallback to manual review/document upload.
- Changing government formats/compliance rules: Modular validation logic for adaptability.
- Ensuring consistent trust scoring despite data variability.

- User error/confusion: Streamlined, guided UI with contextual help throughout.

Milestones & Sequencing

Project Estimate

- Medium project: ~10 weeks total — MVP (4 weeks), Phase 2 (6 weeks).

Team Size & Composition

- Lean startup model:
 - Product Lead (part-time, also acting as QA)
 - 2 Full-Stack Engineers (React/Node)
 - 1 UI/UX Designer (part-time)
 - 1 Data/Integration Specialist (contract as needed)
 - 1 Support/Admin Ops (part-time)
- Total: 3–4 core people

Suggested Phases

Phase 1: MVP Launch (4 weeks)

- Key Deliverables (Product Owner + Engineers + Designer):
 - Vendor onboarding form and validations
 - GSTIN and Udyam/MSME verification integrations
 - Admin dashboard (basic: monitoring/status/risk flags)
 - Core trust score/PDF report
 - Responsive front-end for web/mobile
- Dependencies: API access (GSTIN/Udyam), simple onboarding authentication

Phase 2: Premium Modules & Analytics (6 weeks)

- Key Deliverables (All Core Team + Data Specialist):
 - Director due diligence (MCA21/ZAUBA), payment history (CRIF/Experian)
 - Enhanced analytics/reporting dashboards
 - Advanced admin controls (manual override, case management)
 - User notifications/in-app messaging

- Performance/scale optimization & security enhancements
- Dependencies: Third-party API agreements, premium payment processing